Measure sound speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Analyze sound travel measurements and their limitations.

SUCCESS CRITERIA - I can...

1. Define one-way versus round-trip paths.
2. Calculate speed from distance and time.
3. Use repeated measurements and discuss timing uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “echo” to explain your model or result. Sentence starters are optional.

trial: one run or repetition of a test

echo: a sound received after reflection

round trip: travel from start to a reflector and back

01 Warm-up: bring back an earlier idea

Three instantaneous displacements are +3 mm, -5 mm and -7 mm.
Find their sum and explain why one displacement value does not determine total wave energy.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

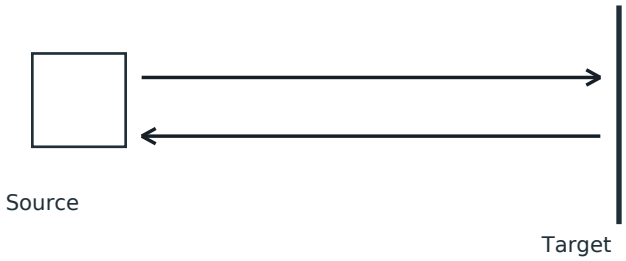
02 Worked example: explain the model

Reference: $v = d/\Delta t$; $v = 2\ d/\Delta t$ for an echo.
Match the distance to one-way or round-trip travel. Use moderate sound levels and record actual data separately.

Worked reasoning: A sound echo returns 0.4 s after a pulse. At 340 m/s, round-trip distance is 136 m, so the reflector is 68 m away.

Your turn: Draw outgoing and returning paths and mark where the factor of 2 enters.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A pulse travels between two sensors separated by 1.70 m. Delay is 0.0050 s. Is this one-way or echo distance?
Find speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

04 Apply the physics

A microphone detects an echo 0.0200 s after a pulse hits the air. The reflecting wall is 3.40 m away.
Find sound speed. Treat distance as exact and timing as 0.0200 ± 0.0005 s; calculate the lowest and highest possible speeds.
Label the round trip.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

Plan a table for three trials. Include distance, elapsed time, method (one-way or echo), speed and one possible systematic error.

Sentence starter: The step I would change is ... . The evidence is ... .

Marzano check - circle: 1 2 3

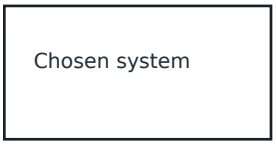
06 Review: connect an earlier idea

A sphere’s radius becomes 4 times as large at fixed density. Derive the area-to-mass multiplier.
Explain why kinetic energy per mass alone does not predict survival during entry.
Label energy entering, leaving and stored.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: I predict ... . My model supports this because ... .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Two sensors 0.85 m apart record a wave’s travel time as 0.0025 s.
Find the wave speed. A timing fault adds 0.001 s to the recorded interval.
Calculate the speed this faulty measurement would give, and explain the direction of the error.
Label the one-way path.

Sentence starter: The step I would change is … . The evidence is … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A microphone pair separated 1.5 m measures travel-time delay 0.0044 s.
Find speed. If delay is uncertain by±0.0002 s, find worst-case speed bounds.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3